VOLODYMYR POLISHCHUK

Head of Software & Autonomy | Systems Architect | GPS-Denied UAV Navigation

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SUMMARY

Systems architect and engineering leader with 19 years of building production systems at scale, from Wix's developer platform to Playtech's real-time betting engine and UkrCard's payment network. Since 2026 I have led software and autonomy at Applied Technology Collective, a Ukrainian UAV maker, where I designed the GPS-denied navigation its drones fly on, along with the flight-controller firmware and simulator behind it. I reason from first principles, breaking a problem down to its physics, mathematics and computing limits rather than copying existing frameworks. I work AI-native: I own the architecture, the mathematics and the acceptance criteria, and AI coding agents are my implementation team.

EXPERIENCE

Applied Technology Collective | Head of Software & Autonomy

Kyiv, Ukraine | 2026 – Present

Ukrainian developer of strike, loitering-munition and cargo UAVs. I joined as the only software engineer, built the software and autonomy team from scratch and hired its first two engineers. NADIR, SEXTANT and ORRERY are my own IP, used by the company; code walkthrough on request.

NADIR — Visual-Inertial Odometry

- Built GPS-denied navigation for the company's drones using visual-inertial odometry from one downward camera, so each drone holds position and flies in guided mode without satellites, on $15 of added hardware.

SEXTANT — Long-Range Terrain Navigation

- Extended GPS-denied flight to long range by matching the shape of the surrounding terrain against a terrain model for absolute position and heading, so navigation keeps working in snow, foliage and changing light, where matching camera images fails.

ORRERY — Standalone Flight Simulator

- Built a flight simulator that runs the real ArduPilot flight code, in which NADIR and SEXTANT must pass their acceptance tests before they reach a real drone, cutting the cost and risk of flight testing.

Wix.com | Principal Software Engineer / Architect

Kyiv, Ukraine | 2017 – 2026

Platformization & Engineering Culture

- Led company-wide adoption of a single API standard, enforced automatically for every service, laying the groundwork for opening Wix's internal APIs to third-party developers and the App Market.
- Moved Wix's backend to asynchronous programming through mandatory courses and workshops I designed and taught for its 1,000+ server engineers. The shift cut the backend's operational resource costs and let its microservices scale further.
- Interviewed 300+ engineering candidates and voted on Wix's hiring committees, setting the bar for who joined engineering.

Platforms & Infrastructure

- Architected Wix's internal serverless platform with a core team of five, freeing about 3,000 developers across 2,000+ microservices from managing infrastructure.
- Designed an AI chat agent for the Wix App Market that answers shoppers' product questions and handles their orders from the store's own catalogue, so e-commerce owners no longer process them by hand.
- Built Wix's own accredited domain registrar and DNS, making Wix independent of third-party registrars.
- Automated SSL certificates for 8M+ domains, keeping every customer's domain secure with no manual renewals.
- Led the migration of domain management from a legacy monolith to microservices, stabilizing the system behind 8M+ domains.

Playtech | Software Architect / Tech Lead

Kyiv, Ukraine | 2012 – 2017

- Led a team of five as tech lead and moved the sports-betting platform from a monolith to a reactive architecture that carried FIFA World Cup peaks of 60M requests per minute.
- Redesigned the deployment to stay highly available on as few as three servers, cutting hardware costs.
- Designed a distributed database running on the company's own servers, keeping the platform within regulation that ruled out the cloud.

UkrCard | Systems Architect

Kyiv, Ukraine | 2007 – 2012

- Led the move of every ATM, POS and self-service terminal from legacy wired lines to encrypted wireless connections, securing the payment network.
- Built a neural-network assistant for fraud detection that learned from operator feedback, cutting costly false positives (Master's project).

SKILLS

Autonomy: visual-inertial navigation, sensor fusion, terrain-referenced navigation, ArduPilot, MAVLink.
Engineering: Rust, C, Scala, Java, Python; embedded Linux, distributed systems, Kubernetes, gRPC.
AI: RAG, LLM applications, neural networks from first principles.

EDUCATION

M.Sc. in Cybernetics | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics. Thesis: Ensuring the Dependability of Transaction Classification in a Payment System.

B.Sc. in Applied Mathematics | Taras Shevchenko National University of Kyiv, Faculty of Cybernetics

MILITARY & OTHER QUALIFICATIONS

Armed Forces of Ukraine (reserve): Junior Lieutenant, military specialty 708200: operation, tactical deployment and maintenance of air-defense radar systems.
IYT Skipper (International Yacht Training): practical navigation.
Interests: world models for robot navigation (V-JEPA, Genesis); basal cognition (Michael Levin); foundations of physics.